RE-MESH

IoT-Based Flood Detection & Disaster Management System

TEAM MEMBERS

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Problem Statement & Stakeholders

Context:

- IoT-Based Flood Detection & Disaster Management System with prediction capabilities

Stakeholders:

- Disaster management authorities & Emergency response teams (Primary)
- Flood-affected communities in low-lying areas (Secondary)

Observed issue:

- During flood situations, there is a lack of early detection, prediction mechanisms, and smart disaster management systems, resulting in delayed response and increased risk.

Why it matters:

- Enables early flood detection and prediction
- Provides real-time monitoring and situational awareness
- Ensures alerts even when traditional networks fail
- Improves response time and community safety

Research Gap:

- Problem: Costly, network-dependent flood systems fail in emergencies.
- Solution: Scalable, low-cost IoT system with real-time sensors, citizen alerts, and dashboards.
- Key Features: real-time monitoring, local/cloud prediction, multi-channel alerts, participatory reporting.
- Reliability: Operates when traditional networks are unavailable.
- Benefits: Cost-effective, fast warnings, stronger community resilience.

IoT-Based Flood Detection & Disaster Management System

Sub-Objectives:

- **IT22196088** - Distributed IoT Nodes & power management
- **IT22571298** - LoRa Routing, Encryption & Compression
- **IT22563446** - ML flood prediction and a node-hosted mini-dashboard.
- **IT22617446** -Main dashboard and management system and IoT-based sound alert integration.

Improvement after PP1

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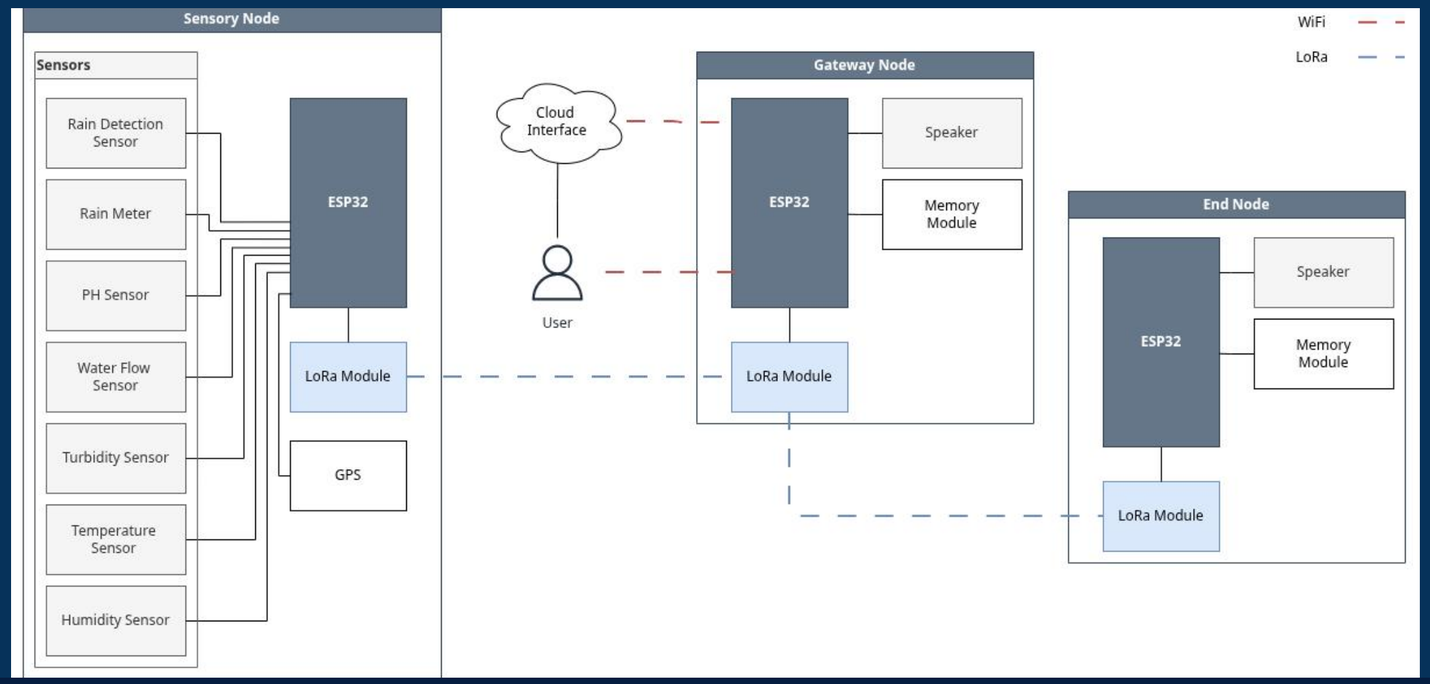
Distributed IoT Nodes & power management

- Assembled all 5 ESP32 nodes with full sensor kits (DHT11, flow, rain×2, turbidity, pH, GPS, INA219)
- Added solar panel + 3-cell LiPo battery pack to each node; verified field power autonomy



Improvement after PP1

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LoRa Routing, Encryption & Compression

- Implemented XOR rolling-counter encryption

  (Encrypted[i] = Data[i] ⊕ Key ⊕ Counter);

  Rolling counter prevents replay attacks
- Compact binary encoding reduces payload from 600 bytes (JSON) to 70 bytes
- Enables transmission within LoRa payload limits
- Gateway sends complete route trace to Laravel API

Improvement after PP1

Flood Prediction and End-User Dashboard

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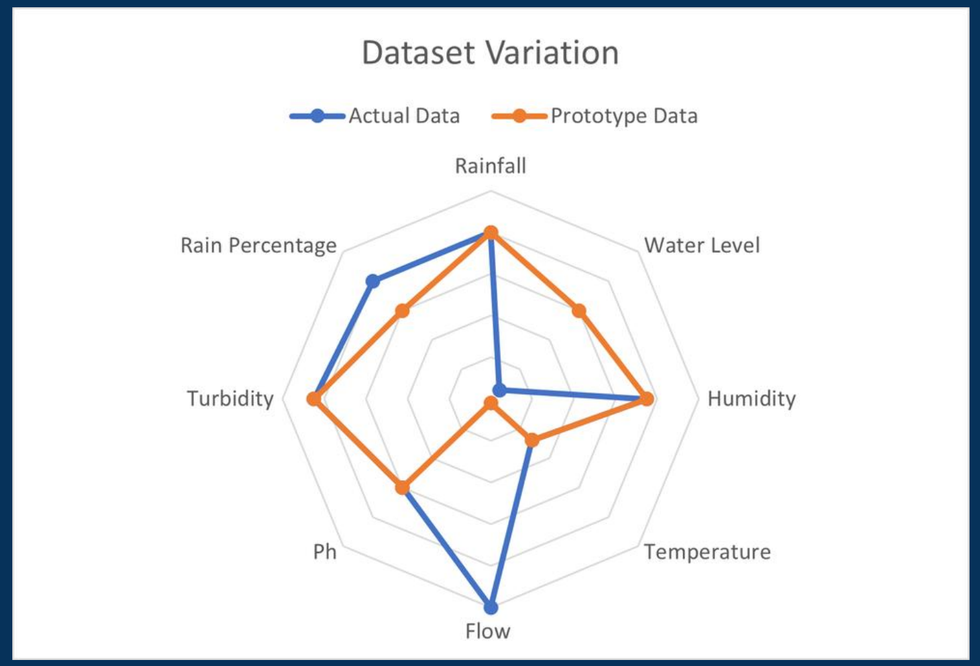
- **Dataset Enhancement** - Expanded to 11 years of historical data, combined Irrigation Department + Kaggle datasets
- **Model Training** - Trained and compared 5 algorithms, Selected best performer based on accuracy
- **OpenWeather API** - Integrated for real-time weather data, Enhances prediction with live rainfall forecasts
- **Offline Dashboard** (emergency access)



Improvement after PP1

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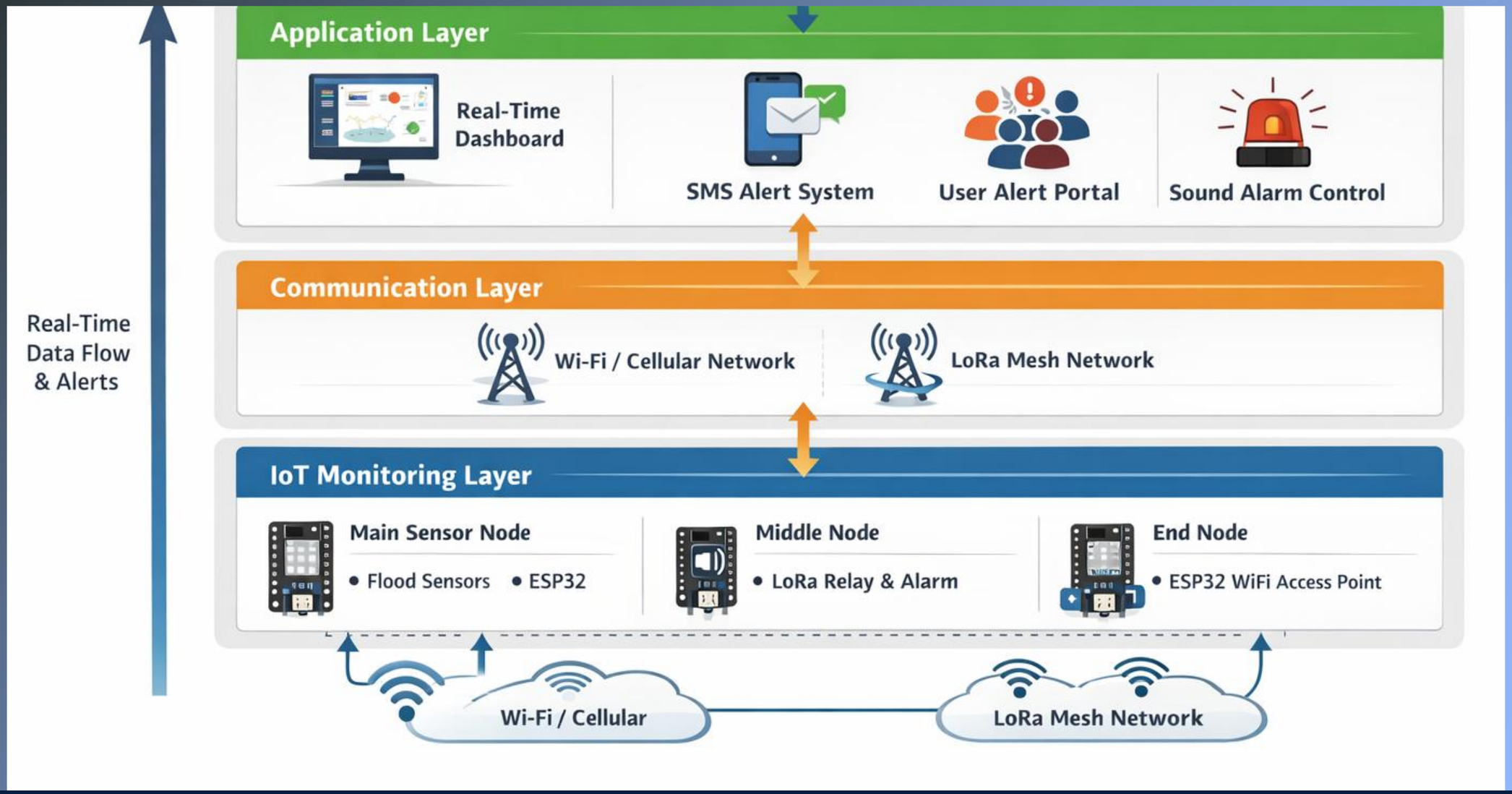
Main dashboard and management system:

- **Real-time Node Monitoring:** - Sensor readings, location, and node health.
- **Sound Alert System:** - Integrated with nodes for local alerts.
- **Log Management:** - Capturing and visualizing node and system logs.
- **Emergency Alert Handling:** - Submission and management of user-reported alerts.
- **SMS Alert Mechanism:** - Sending alerts to predefined groups.
- **User Management:** - Role-based access and control implemented.
- **Rivernet.lk Integration:** - Access to live river data from the main dashboard.

SYSTEM ARCHITECTURE & INTEGRATION



IN PROGRESS

- Designing and fabricating weatherproof enclosures for each sensor node
- Ensuring water resistance and durability for outdoor deployment

COMMERCIALIZATION / IMPACT

Target Users / Market:

Disaster management authorities, emergency response teams, smart city administrators, and IoT solution providers.

Value Proposition:

Low-cost, IoT-ready flood monitoring and disaster management system with real-time alerts, predictive analytics, and offline network support.

Cost and Feasibility:

Affordable deployment using ESP32-based IoT nodes; scalable system architecture with low maintenance and subscription-based analytics for sustainability.

Adoption Plan:

Pilot projects with local authorities → Integration with smart city platforms → Subscription and licensing for broader rollout.

THANK YOU